

What we're doing

How good was the best contacts-v1 model we had on any given date — on contact R-precision and on held-out validation loss — and how tightly do those two track each other?

Every number exists somewhere already (#67, #75, #82, #89, #108, #117, #120, #146, #155, #160, #166, #169), but they are scattered across issue comments, per-experiment CSVs and two W&B projects, with three different inference recipes mixed in. This experiment collects them into one dataset and three standing figures, and keeps them refreshable as new models land.

Why

Not a hypothesis-testing experiment — it is a **standing tracker**. The two things it is built to make visible, both of which prior issues assert piecemeal:

1. Progress on contact accuracy has come almost entirely from the **base model**, not from inference or post-training. 2. Validation loss predicts contact accuracy **across training generations** and stops predicting it **within a generation** (#169's finding), so the loss frontier and the accuracy frontier are not the same curve.

Results so far

The accuracy frontier moved in three jumps, all from the base model, none from inference or post-training: ~0.03 to 0.425 when #75's E8 rung finished (2026-06-21), 0.436 to 0.534 when #117's 16-epoch bs256 run finished (2026-07-22), and 0.534 to 0.562 when #166's amino-acid augmentation continue-train of #117 finished, 2026-07-31. #166 is the first jump from a **data** change rather than a hyperparameter sweep or epoch count.

Between the first two, five weeks of post-training and inference work moved it by +0.011 (#120's re-epoch), and #160's backtracking fine-tune moved it by -0.020.

Two data-side results landed a day apart. #155's crops+contacts-v1+ESM-Atlas 3-way mixture restart finished 08-01 at 0.554 — 0.008 below #166 and a day later, so it never appears as a step. They were not run against each other.

#166 is also the only frontier point with a within-run control: #190 re-scored its own #117 initialization alongside it (0.5336 vs #169's 0.5344), making the +0.0282 a paired result rather than a cross-harness subtraction.

A diagnostic on #155's final checkpoint: scoring each of its 100 sampled rollouts per protein on its own best contacts, instead of voting them together, reads 0.595 — +0.041 of headroom that isn't reachable without ground truth, but points at reranking/selection as an unexplored lever.

Structure predictors on the same 554 proteins: Protenix-v2 single-seq 0.603, ESMFold 0.755, ESMFold2 0.786, Protenix-v2 + MSA 0.812. We are still below all of them, though the single-sequence gap is now 0.041.

Loss tracks accuracy across generations, not inside one

The 0.053-nat #75 to #117 gap buys +0.109 R-precision — about 2 R-precision per nat. The 0.008-nat gap between #117's early-stop and final checkpoints buys nothing, and #146's 3B is 0.0012 better on loss and 0.023 worse on R-precision.

So the loss frontier is useful for deciding which checkpoints are worth scoring, and useless for picking between two checkpoints of one run.

The trap in these numbers

The same weights score ~ 0.086 higher under exp82's rollout recipe than under exp89's original pairwise scorer (#61/#75 E8: 0.339 to 0.425; #120: 0.350 to 0.436). That is comparable to two generations of model progress.

The figures keep pairwise and rollout visually separate. Anything from the eval-checkpoint skill is pairwise; anything from exp82's rollout workers or the exp169 dispatcher is rollout. Never infer the recipe from the magnitude.

A third recipe, oracle best-of-100, is a diagnostic upper bound, not a deployable score (see Results so far) — it never enters the "best model trained to date" frontier line or headline labelling, only its own marker.

Per-protein comparison with Protenix-v2

Over the 554 proteins, Marifold #166 scores 0.562. Protenix-v2 single-sequence scores 0.603 (paired difference -0.041, Marifold higher on 34%). Protenix-v2 with MSAs scores 0.812 (paired difference -0.250, Marifold higher on 8%).

Re-pointed from #117 to #166 the shape did not change, only the gap: every length bin improved and the single-sequence deficit halved.

The two baselines trend opposite ways with length

Against single-sequence Protenix the gap narrows with length and changes sign in the > 400 bin: -0.107 below 100 residues, +0.122 above 400. The 200-400 bin is now -0.011, which on 171 proteins is a tie.

Against MSA Protenix it widens: -0.195 below 100 residues, -0.471 above 400. MarFold does not win a single protein above 400 residues in that comparison.

The reason is in the marginals. MarFold declines with length (0.56 to 0.39) and single-sequence Protenix declines faster (0.66 to 0.27), while MSA Protenix improves with length (0.75 to 0.86).

So "MarFold holds up better on long proteins" is about the single-sequence baseline only, it is a shallower decline rather than absolute strength, and that bin holds 17 proteins either way.

Open gap

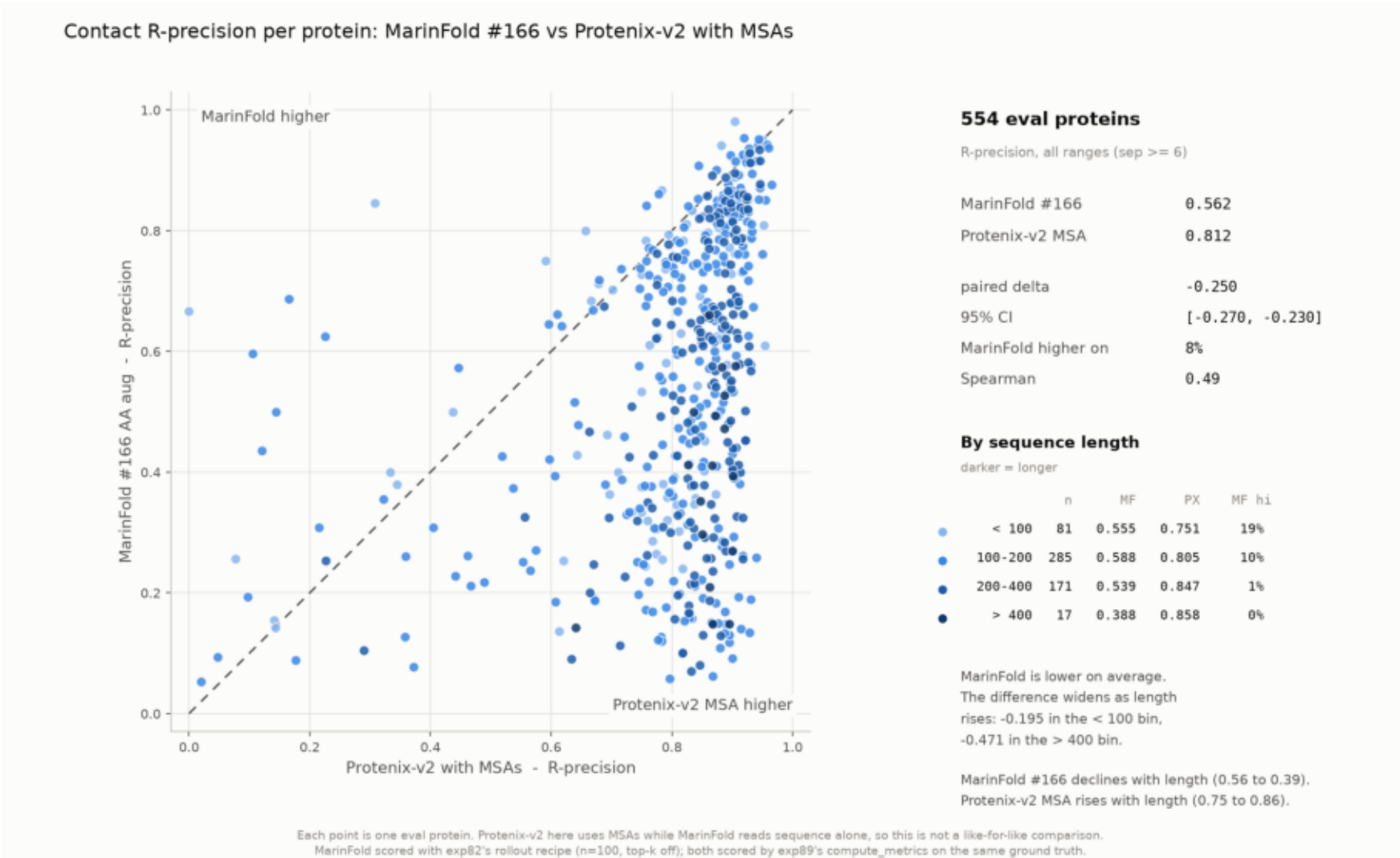
#108's 3B on CoreWeave held the loss frontier from 2026-07-11 to 07-16 at 2.7418 and was never contact-scored. Given #146 — a 3B at matched loss under-performing the 1.5B — it is probably not a missed frontier point, but that is an inference, not a measurement.

Keeping this current

Two commands: `build_dataset.py` re-pulls W&B, `plot_progress.py` redraws (plus `plot_vs_protenix.py` for the head-to-head pair). New benchmark scores are hand-added to `RPRECISION_ROWS` with a source citation and an explicit inference recipe; a new W&B sweep tag goes in `WANDB_SOURCES`. When a new model takes the frontier, re-point `MARINFOLD_MODEL` and `ROWS_CSV` in `plot_vs_protenix.py` — it needs published per-protein rows, not a summary. Full procedure in the `README`.

marifold_vs_protenix_msa.png

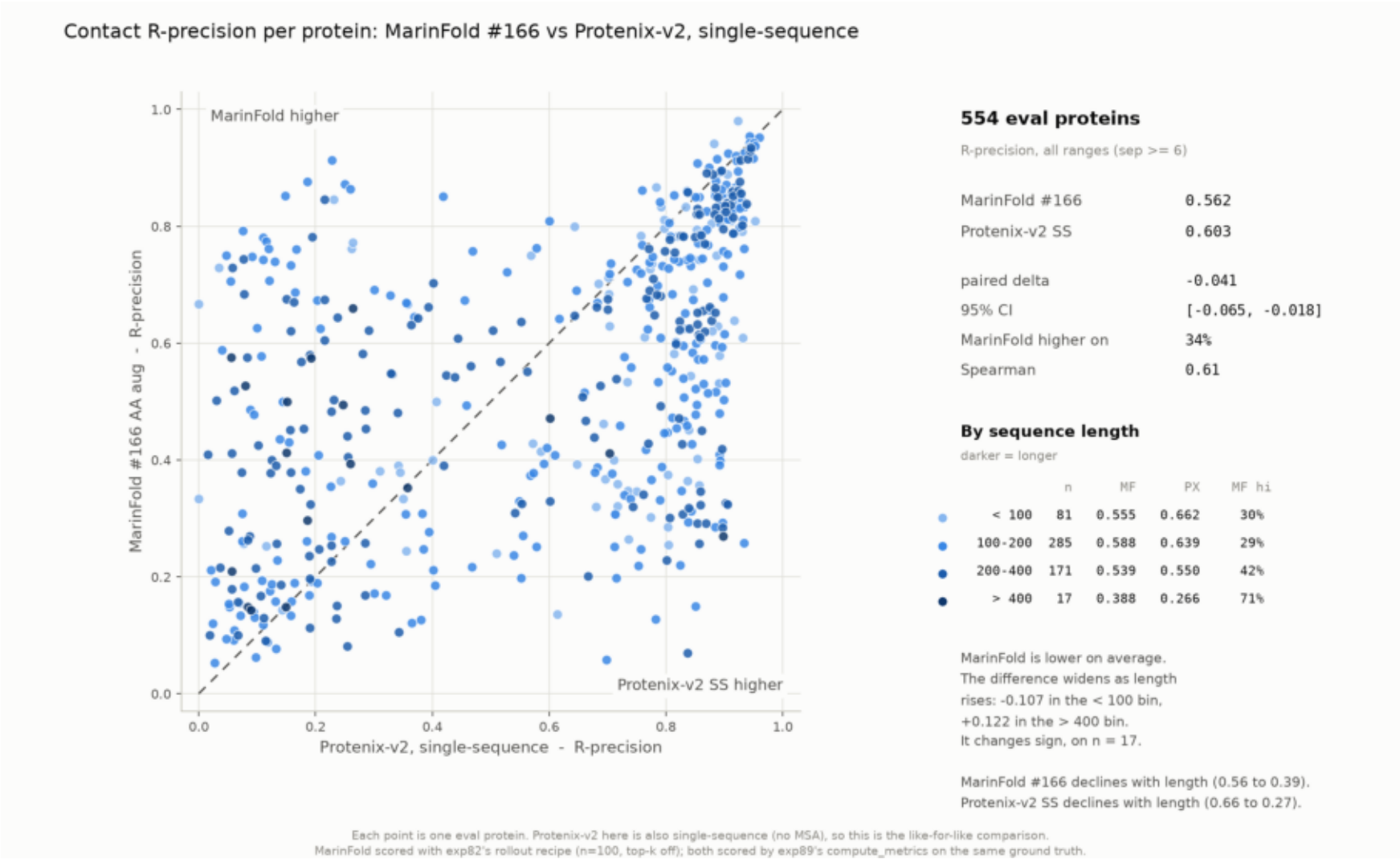
Per-protein R-precision, Marinfold #166 vs Protenix-v2 MSA (n=554). Means 0.562 vs 0.812; the difference varies with length.



```
$ plot_vs_protenix.py --baseline msa
```

marifold_vs_protenix_ss.png

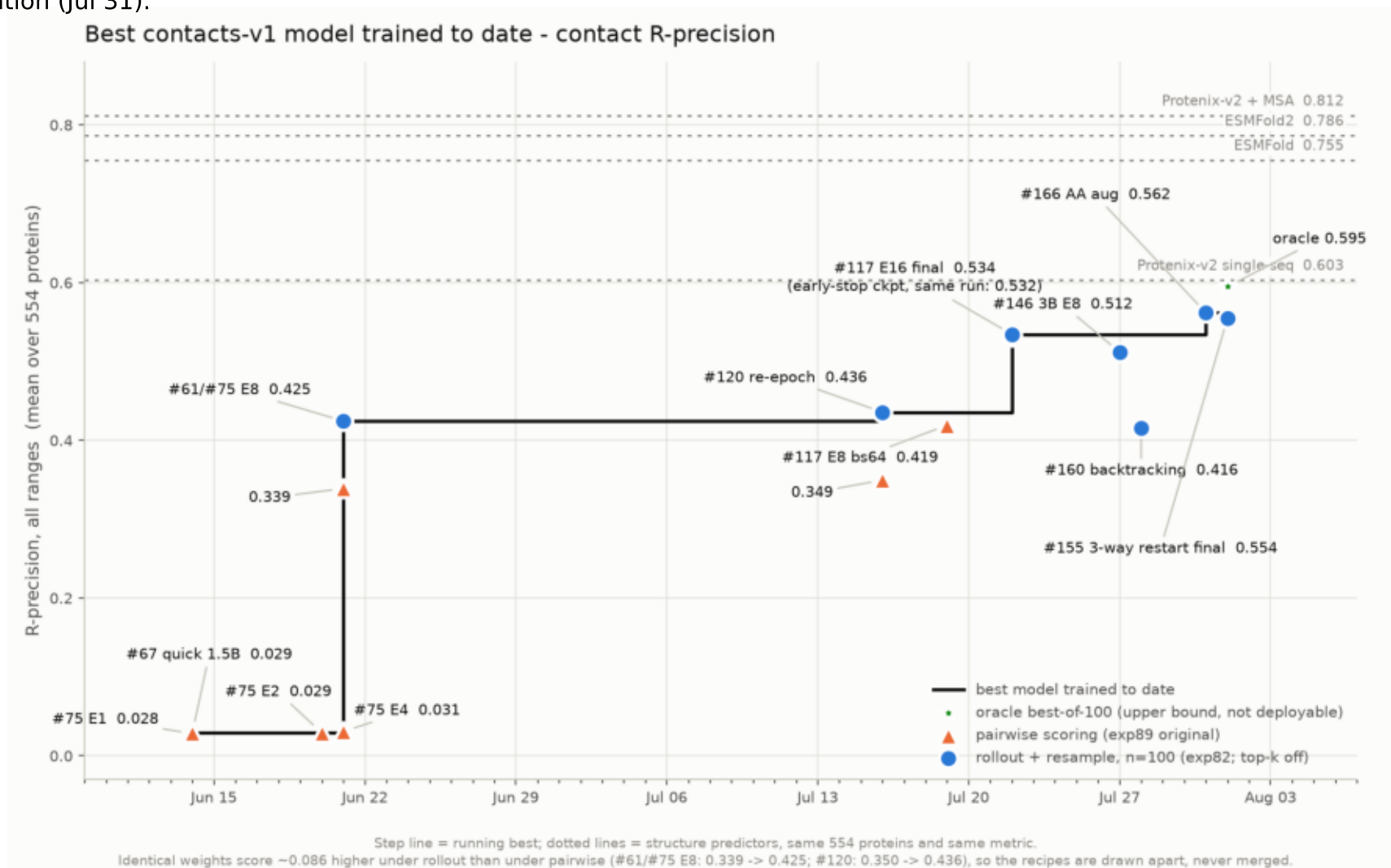
Per-protein R-precision, MarInFold #166 vs Protenix-v2 SS (n=554). Means 0.562 vs 0.603; the difference varies with length.



```
$ plot_vs_protenix.py --baseline single_seq
```

rprecision_frontier.png

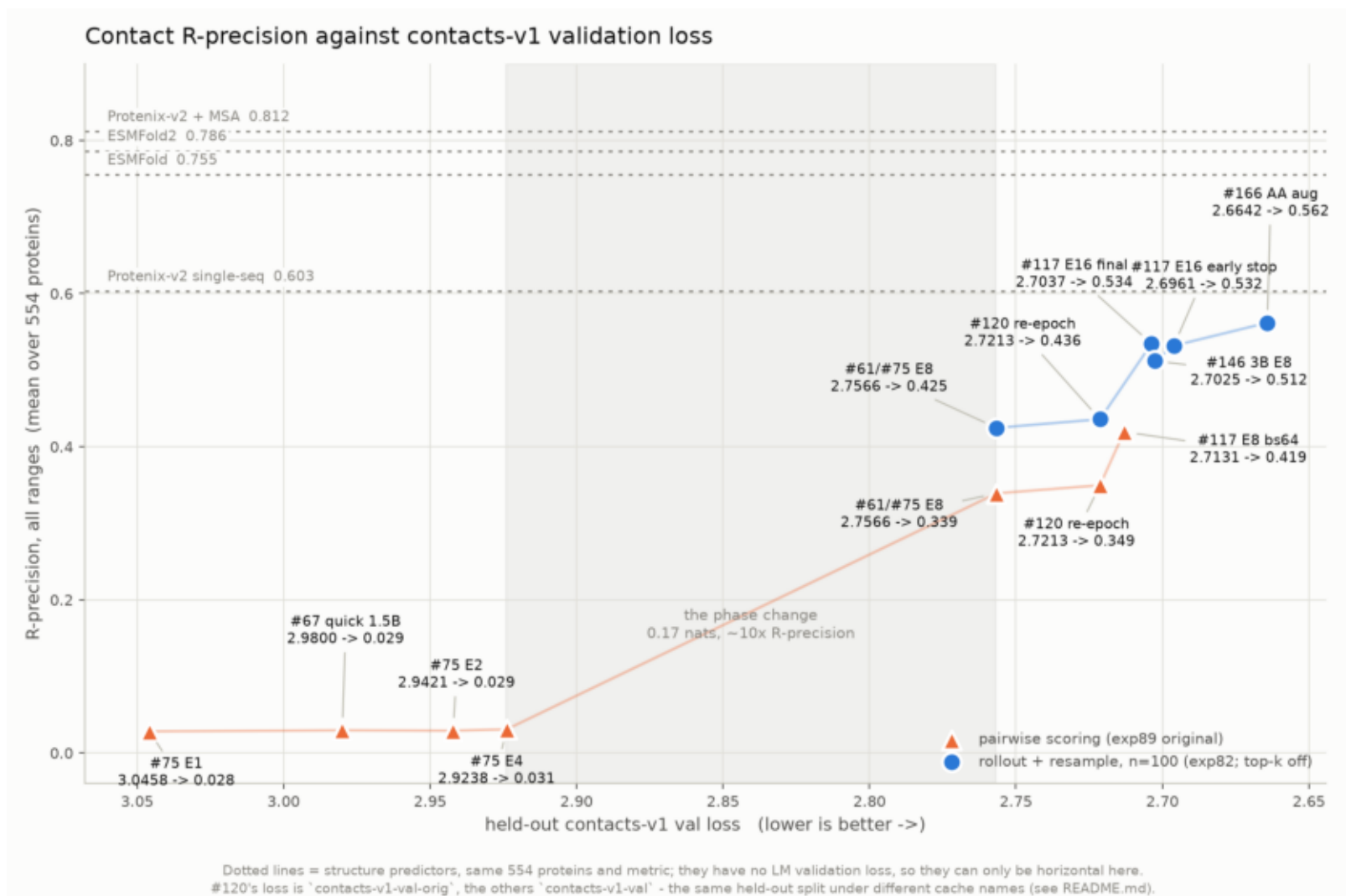
Best contacts-v1 model to date on contact R-precision. Three jumps, all from the base model: #75 E8 (Jun 21), #117 E16 (Jul 22), #166 AA augmentation (Jul 31).



\$ plot_progress.py

rprecision_vs_val_loss.png

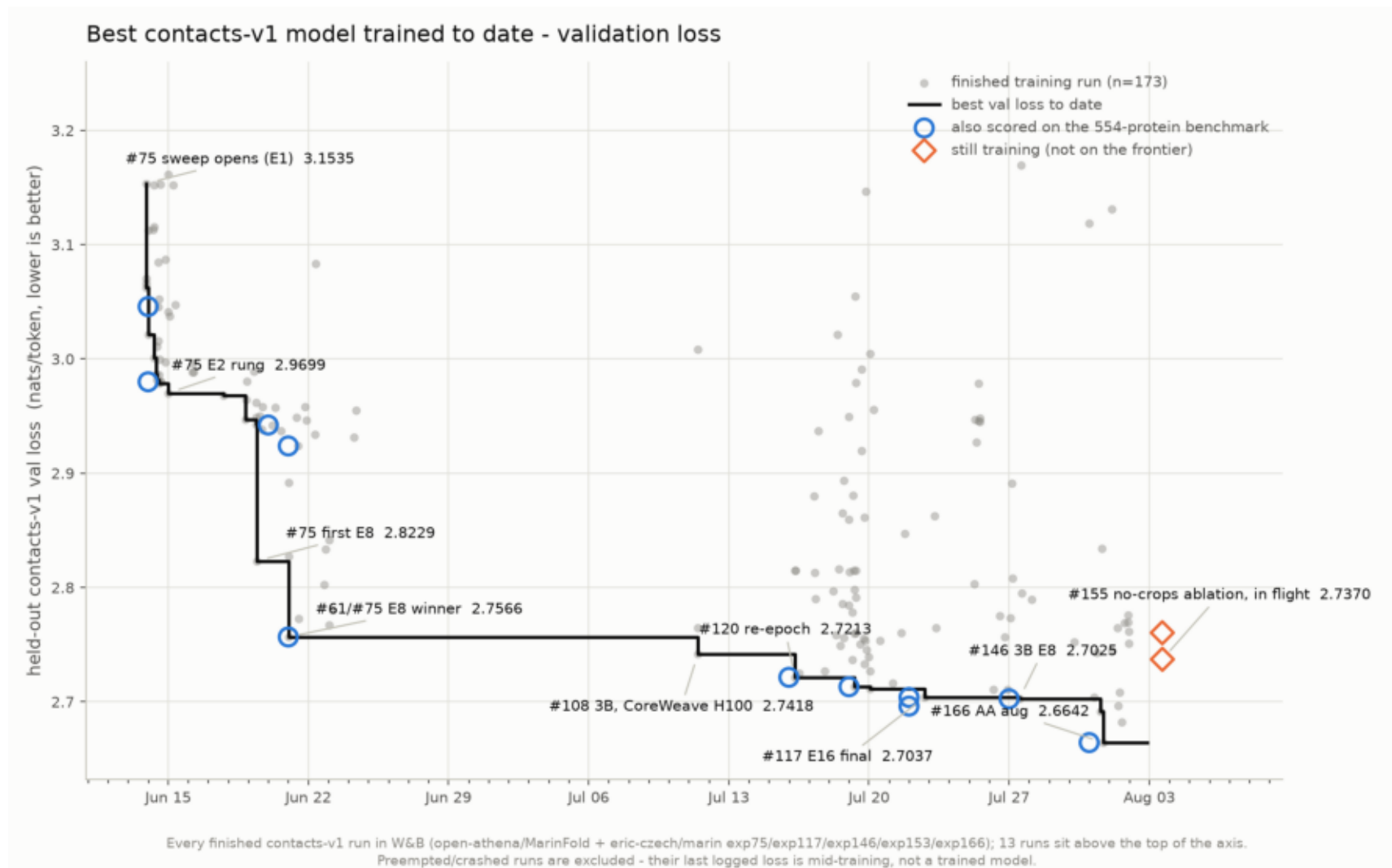
R-precision against val loss. ~2 R-precision per nat across generations; flat inside one run (#117 early vs final).



\$ plot_progress.py

val_loss_frontier.png

Best held-out contacts-v1 val loss to date over 173 finished runs. The loss frontier has many more steps than the accuracy one.



\$ plot_progress.py